University of Asia Pacific

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Third Year, Second Semester

Course Name: Software Engineering Lab

Course Code: CSE-314

Date of Submission: 18/02/2026

Group No: 02

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Athletix

(Student Intelligence for Athletic Fitness & Health)

1. Problem Statement

In many educational institutions, student athletes follow generalized training programs that do not consider individual performance levels, fatigue, recovery status, or health conditions. Coaches often manage multiple athletes simultaneously, making it difficult to closely monitor workload, recovery patterns, and injury risks. Performance tracking is frequently done manually or using disconnected tools, leading to incomplete data and delayed feedback.

Moreover, students have limited access to medical professionals for early reporting of fatigue, pain, or minor injuries. Medical intervention usually occurs only after serious injuries rather than through continuous preventive monitoring. The absence of a centralized system that integrates training data, recovery monitoring, and medical supervision increases injury risk and reduces the effectiveness of athletic development. This creates the need for a precision-based system that supports safe, personalized, and medically guided athletic growth.

Key Issues

- **Lack of Personalized Training Guidance:** Training plans are not adapted to individual performance and recovery conditions
- **Manual Performance Tracking:** Data collection is time-consuming and prone to errors
- **No Centralized Platform:** Training, recovery, and medical data are scattered across tools
- **Limited Coach Visibility:** Coaches cannot easily track fatigue, recovery, or injury trends
- **Delayed Medical Support:** Students lack early access to medical staff for symptom reporting
- **High Injury Risk:** Overtraining and poor recovery monitoring increase injury probability

2. Proposed Solution

Athletix is a student-centric digital platform that integrates athletic performance tracking, fatigue and recovery monitoring, and medical supervision into a single unified system. The platform analysis training activity, workload, fatigue indicators, and recovery trends to provide precision-based training and recovery recommendations.

The system enables structured collaboration among students, coaches, and medical personnel. Medical staff can monitor reported symptoms and injury risk indicators to provide early guidance and preventive care. This data-driven and medically informed approach ensures personalized, safe, and continuously updated athletic development for students.

Core Features

- Student performance tracking (speed, distance, intensity)
- Fatigue and recovery analysis
- Precision-based training recommendations
- Coach and medical staff assignment
- Student–medical staff messaging for health concerns
- Injury risk alerts and preventive recommendations
- Progress visualization through interactive dashboards

3. CEP (Complex Engineering Problem) Justification

Integrates athletic performance analytics with health monitoring under uncertain and variable conditions. Requires balancing performance improvement with injury prevention and medical safety. Supports collaboration among multiple stakeholders including students, coaches, and medical staff ·Uses historical and real-time performance and health data for decision-making. Involves uncertainty in athlete workload, fatigue, recovery, and injury risk assessment. Incorrect system decisions may have direct health and safety consequences for student athletes

Complex Engineering Problem Attributes (P)

- **P1:** Requires in-depth engineering knowledge in data analysis and system architecture
- **P2:** Involves conflicting requirements between training intensity and medical safety
- **P3:** Contains complex relationships between workload, fatigue, recovery, and injury risk
- **P5:** Has significant health and injury-prevention consequences
- **P6:** Involves multiple stakeholders including students, coaches, medical staff, and administrators

Complex Engineering Activities (A)

- **A1:** Integration of athletes, coaches, medical staff, and performance data
- **A2:** Trade-off analysis between athletic performance optimization and health safety
- **A3:** Creative design of a precision training advisory and medical alert engine
- **A4:** Positive societal impact on student health, safety, and sports performance
- **A5:** Development of a novel academic solution beyond traditional fitness tracking systems

Knowledge Profile (K)

- **K1:** Human physiology and sports medicine for fatigue, recovery, and injury modeling
- **K2:** Mathematics and statistics for performance analysis and injury risk prediction
- **K3:** Engineering fundamentals for secure and reliable software architecture
- **K4:** Specialist knowledge in precision training and medical alert systems
- **K5:** Engineering design knowledge for dashboards and performance visualization
- **K6:** Engineering practice including software development and database management
- **K7:** Ethical responsibility and student safety awareness in health-related systems
- **K8:** Engagement with research literature in sports science and medical monitoring

4. Similar Applications

- **Strava** – Activity tracking only
- **Nike Training Club** – Workout guidance without analytics
- **Garmin Connect** – Hardware-dependent performance tracking
- **Google Fit** – General health tracking without precision advisory or medical focus

5. Conclusion

Athletix provides a centralized, preventive, and data-driven solution for managing student athletic performance and health. By integrating training analytics, recovery monitoring, and medical supervision into a single platform, the system reduces injury risks while improving athletic outcomes. Its multi-stakeholder design supports early intervention and personalized guidance. From an engineering perspective, Athletix clearly addresses a Complex Engineering Problem involving uncertainty, conflicting requirements, and health-related consequences, making it a strong academic project with real-world relevance and societal impact.

Feasibility Analysis

The feasibility analysis evaluates whether the Athletix project can be successfully developed, implemented, and sustained within realistic technical, operational, economic, and legal constraints. Based on the project scope and objectives, Athletix is found to be **feasible and practical** as an academic and real-world system.

Technical Feasibility

The system will be built using a robust and scalable "Full-Stack" approach to ensure fast development and high security:

- **Backend: Python** with the **Django** framework (chosen for its built-in security and rapid development features).
- **Frontend: HTML5**, **CSS3**, and **JavaScript** (to create responsive, interactive dashboards for mobile and desktop).
- **Database: SQLite** (to handle complex relational data between students, coaches, and medics).
- **Hosting:** Budget-friendly host platform

Operational Feasibility

Athletix fits well into existing academic and sports environments.

- Students can easily input training data and report symptoms through a user-friendly interface.
- Coaches gain centralized visibility of athlete performance, workload, and recovery trends, reducing manual effort.
- Medical staff can monitor health indicators remotely and provide early preventive guidance.
- The platform supports collaboration among multiple stakeholders without disrupting existing training routines.
- Minimal training is required for users due to intuitive dashboards and structured workflows.

Economic Feasibility

The project is financially viable, with the budget allocated to ensure high-quality software engineering, data security for medical records, and institutional-grade hosting.

Category	Details	Budget (BDT)
Infrastructure	High-performance VPS (Cloud), Domain (.com.bd), SSL Certificates, and Secure Data Backups.	60,000
Product Development	Full-stack development using **Python/Django**, custom **JavaScript** analytics, and **SQLite** optimization.	2,50,000
UI/UX Design	Professional, responsive web design for Athlete, Coach, and Medical Staff dashboards.	40,000
Security & Testing	Penetration testing, security audits for medical data, and bug fixing.	30,000
Operations	Stakeholder training, pilot launch support, and final documentation.	20,000
Total Startup Capital		**4,00,000**

Schedule Feasibility

The schedule is optimized for a full web-based rollout, moving from initial requirements to a live pilot launch.

Week	Phase	Key Task / Deliverable
Week 1	Requirement Analysis & Planning	Stakeholder interviews & user "pain point" mapping.
Week 2	System Design (Architecture & UI/UX)	Wireframing, ER Diagrams, and Django project initialization.
Week 3–4	Backend Development	Setting up **SQLite** models, User Auth, and Core API logic.
Week 5–6	Web Interface Development	Building the responsive athlete forms for pain and intensity logging.
Week 7	Web Dashboard Development	Creating the Coach and Medical staff analytics portals.
Week 8	Analytics & Injury Risk Module	Coding the **Python** logic for fatigue scores and risk alerts.
Week 9	Integration & System Testing	Unit testing, bug fixing, and security/encryption audits.
Week 10	Final Deployment & Documentation	Hosting on VPS, pilot launch, and final report submission.

Legal and Ethical Feasibility

Athletix is legally and ethically feasible when appropriate safeguards are implemented throughout system design and operation. The platform functions as a decision-support system rather than a medical diagnostic tool, which helps minimize legal liability. User privacy and data protection are ensured through secure authentication, role-based authorization, and encrypted data storage and transmission. Clear user consent mechanisms can be applied for the collection and analysis of health-related data, ensuring transparency and compliance with institutional policies. Ethical engineering principles are upheld by prioritizing student safety, responsible data usage, and the prevention of harm, making the system suitable for deployment within academic and sports institutions.

Social and Educational Feasibility

Athletix demonstrates strong social and educational feasibility by positively contributing to student health, safety, and learning outcomes. The system promotes injury prevention and responsible training practices among student athletes through continuous monitoring and early risk identification. It encourages data-driven decision-making in sports education by providing actionable insights to students, coaches, and medical staff. Athletix also enhances collaboration between educational institutions and sports medicine professionals, fostering a multidisciplinary approach to athletic development. From an academic perspective, the platform serves as a valuable learning tool for students in engineering, data analysis, and health informatics, thereby supporting both educational objectives and societal well-being..

Conclusion

Based on technical, operational, economic, legal, and social considerations, **Athletix is a highly feasible project**. It addresses a real-world problem using practical engineering solutions while aligning with academic objectives and ethical responsibilities. The system is suitable as a **Complex Engineering Problem–based academic project** with strong potential for real-life adoption and future enhancement.

Role

1.Project Manager - 23101219

2. System Analyst / Business Analyst - 23101207

3. System Designer / Software Architect - 23101194

4. Software Developer (Programmer)- 23101218

5. Tester / Quality Assurance (QA) Engineer - 23101194